Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Electrolysis of dilute aqueous NaCl solution was carried out by passing $10\ \mathrm{mA}$ current. The time required to liberate $0.01\ \mathrm{mol}$ of $\mathrm{H_2}$ gas at the cathode is $\left(1\ \mathrm{F} = 96500\mathrm{C\ mol^{-1}}\right)$

(A) $9.65 \times 10^4\ \mathrm{s}$

(B) $19.3 \times 10^4\ \mathrm{s}$

(C) $28.95 \times 10^4\ \mathrm{s}$

(D) $38.6 \times 10^4\ \mathrm{s}$

Q.2

At $25^\circ\mathrm{C}$, the resistance of a cell filled with $0.01\ \mathrm{M\ KCl}$ solution is $525\ \mathrm{ohms}$. The resistance of the same cell filled with $0.1\ \mathrm{M}\ \mathrm{NH_4OH}$ is $2030\ \mathrm{ohms}$. By calculating the degree of dissociation, calculate the equilibrium constant of $\mathrm{NH_4OH}$. Molar conductivity at infinite dilution for K+ and Cl- is 73.52 and 76.34 respectively.

(A) $2.05 \times 10^{-4}\mathrm{M}$

(B) $2.15 \times 10^{-5}\mathrm{M}$

(C) $1.05 \times 10^{-7}\mathrm{M}$

(D) $2.05 \times 10^{-5}\mathrm{M}$

Q.3

The standard oxidation potentials of $\mathrm{Cu/Cu^{2+}}$ and $\mathrm{Cu^{+}/Cu^{2+}}$ are $-0.34\ \mathrm{V}$ and $-0.16\ \mathrm{V}$ respectively. The standard electrode potential of $\mathrm{Cu^{+}/Cu}$ would be

(A) $0.18\ \mathrm{V}$

(B) $0.52\ \mathrm{V}$

(C) $0.82\ \mathrm{V}$

(D) $0.49\ \mathrm{V}$

Q.4

A current of $0.250\ \mathrm{A}$ is passed through $400\ \mathrm{ml}$ of a $2.0\ \mathrm{M}$ solution of NaCl for $35\mathrm{minutes}$. What will be the pH of the solution after the current is turned off ?

(A) 12.98

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(B) 12.13

(C) 10.48

(D) 9.24

Q.5

How much charge should be supplied to a cell for the electrolytic production of $245\ \mathrm{gm}\ \mathrm{NaClO_4}$ from $\mathrm{NaClO_3}$ if the anode efficiency for the required reaction is 60% ?

(A) $6.43 \times 10^5 \mathrm{C}$

(B) $6.67\ \mathrm{F}$

(C) $6.43 \times 10^6 \mathrm{C}$

(D) $66.67\ \mathrm{F}$

Q.6

Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration.
Some half cell reactions and their standard potentials are given below

$\mathrm{MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightarrow Mn^{+2}(aq) + 4H_2O(l); E^\circ = 1.51\ V}$

$\mathrm{Cr_2O_7{}^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{+3}(aq) + 7H_2O(l); E^\circ = 1.38\ V}$

$\mathrm{Fe^{+3}(aq) + e^- \rightarrow Fe^{+2}; E^0 = 0.77\ V}$

$\mathrm{Cl_2(\ g) + 2e^- \rightarrow 2Cl^-(aq); E^0 = 1.40\ V}$

Identify the only incorrect statement regarding the quantitative estimation of aqueous $\mathrm{Fe\left(NO_3\right)_2}$

(A) $\mathrm{MnO_4^-}$ can be used in aqueous HCl

(B) $\mathrm{Cr_2O_7{}^{2-}}$ can be used in aqueous HCl

(C) $\mathrm{MnO_4^-}$ can be used in aqueous $\mathrm{H_2SO_4}$

(D) $\mathrm{Cr_2O_7{}^{2-}}$ can be used in aqueous $\mathrm{H_2SO_4}$

Q.7

The molar conductivities $\wedge^{\circ}_{\mathrm{NaOAc}}$ and $\wedge^{\circ}_{\mathrm{HCl}}$ at infinite dilution in water at $25^{\circ}\mathrm{C}$ are 91.0 and $426.2\ \mathrm{S\ cm^2/mol}$ respectively. To calculate $\wedge^{\circ}{}_{\mathrm{HOAc}}$, the additional value required is

(A) $\wedge^{\circ}{}_{\mathrm{NaCl}}$

(B) $\wedge^{\circ}{}_{\mathrm{H_2O}}$

(C) $\wedge^{\circ}{}_{\mathrm{KCl}}$

(D) $\wedge^{\circ}{}_{\mathrm{NaOH}}$

Q.8

Which of the following is a highly corrosive salt

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(A) $FeCl_2$

(B) $PbCl_2$

(C) Hg_2Cl_2

(D) $HgCl_2$

Q.9

The spontaneous redox reaction/s among the following is/are

(1) $2Fe^{3+} + Fe \rightarrow 3Fe^{++}$

(2) $Hg_2^{++} \rightarrow Hg^{++} + Hg$

(3) $3AgCl + NO + 2H_2O \rightarrow 3Ag + 3Cl^- + NO_3^- + 4H^+$

Given that

$E^0_{Fe^{+++}/Fe^{++}} = 0.77\ V, E^0_{Fe^{++}/Fe} = -0.44\ V$

$E^0_{Hg_2^{++}/Hg} = 0.85\ V, \quad E^o_{Hg^{++}/Hg_2{}^{++}} = 0.92\ V$

$E^o_{AgCl/Ag} = 0.22\ V, \quad E^0_{NO_3^-/NO} = 0.96\ V$

(A) 1

(B) $1, 2, 3$

(C) $1, 2$

(D) $1, 3$

Q.10

Which one of the following metals could not be obtained on electrolysis of aqueous solution of its salts

(A) Ag

(B) Mg

(C) Cu

(D) Cr

Q.11

Cost of electricity for the production of x L H_2 at STP at cathode is Rs. x, then cost of electricity for the production of x L O_2 gas at STP at anode will be (assume 1 mol of electrons as one unit electricity) :

(A) $2x$

(B) $4x$

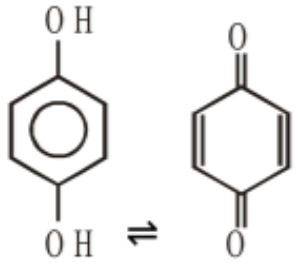
(C) $16x$

(D) $32x$

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Q.12

For the half-cell



$+2H^{+}+2e^{-} \quad E^{0}=1.30\ V$

At $pH=2$, electrode potential is :

(A) $1.36\ V$

(B) $1.30\ V$

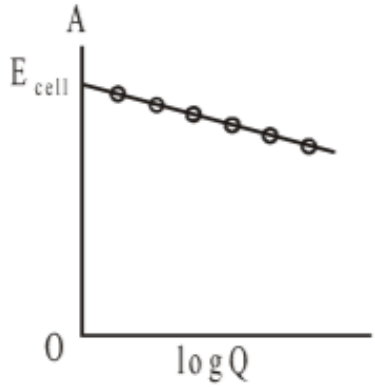
(C) $1.42\ V$

(D) $1.20\ V$

Q.13

$Zn+Cu^{2+}(aq) \rightleftharpoons Cu+Zn^{2+}(aq)$

Reaction quotient is $Q=\frac{[Zn^{2+}]}{[Cu^{2+}]}$



Variation of E_{cell} with $\log Q$ is of the type with $OA=1.10\ V.E_{cell}$ will be $1.1591\ V$ when :

(A) $\frac{[Cu^{2+}]}{[Zn^{2+}]}=0.01$

(B) $\frac{[Zn^{2+}]}{[Cu^{2+}]}=0.01$

(C) $\frac{[Zn^{2+}]}{[Cu^{2+}]}=0.1$

(D) $\frac{[Zn^{2+}]}{[Cu^{2+}]}=1$

Q.14

The standard electrode potential is measured by

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(A) Electrometer

(B) Voltmeter

(C) Potentiometer

(D) Galvanometer

Q.15

"Maintenance free" batteries now in use in place of common batteries have

(A) Electrode made of lead-lead oxide

(B) Electrodes made of calcium-containing lead alloy

(C) Non aqueous solvents as medium

(D) Platinum electrodes

Q.16

Saturated solution of KNO_3 is used to make 'salt-bridge' because

(A) velocity of K^+ is greater than that of NO_3^-

(B) velocity of NO_3^- is greater than that of K^+

(C) velocity of both K^+ and NO_3^- are nearly the same

(D) KNO_3 is highly soluble in water

Q.17

Which of the following reaction is possible at anode

(A) $2Cr^{3+} + 7H_2O \rightarrow Cr_2O_7^{2-} + 14H^+$

(B) $F_2 \rightarrow 2F^-$

(C) $\frac{1}{2}O_2 + 2H^+ \rightarrow H_2O$

(D) None of these

Q.18

The reaction
$\frac{1}{2}H_2(\ g) + AgCl(s) \rightarrow H^+(aq) + Cl^-(aq) + Ag(s)$
occurs in the galvanic cell

(A) $Ag/AgCl(s)KCl(soln)||AgNO_3(soln)/Ag$

(B) $Pt/H_2(\ g)HCl(soln)||AgNO_3(soln)/Ag$

(C) $Pt/H_2(\ g)HCl(soln)||AgCl(s)/Ag$

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(D) $\mathrm{Pt/H_2(\ g)KCl(soln)||AgCl(}s\mathrm{)/Ag}$

Q.19

A gas X at 1 atm is bubbled through a solution containing a mixture of $1\ \mathrm{MY^-}$ & $1\ \mathrm{MZ^-}$ at $25^\circ\mathrm{C}$. If the reduction potential of $\mathrm{Z} > \mathrm{Y} > \mathrm{X}$, then

(A) Y will oxidize X and not Z

(B) Y will oxidize Z and not X

(C) Y will oxidize both X and Z

(D) Y will reduce both X and Z

Q.20

Find the oxidation potential of a hydrogen electrode at $pH = 10$ and $pH_2 = 1$.

(A) $0.059\ \mathrm{V}$

(B) $0.59\ \mathrm{V}$

(C) $0.00\ \mathrm{V}$

(D) $0.51\ \mathrm{V}$

Q.21

Acidified water is electrolysed using an inert electrode. The volume of gases liberated at STP is $0.168\ \mathrm{L}$. The quantity of charge passed through the acidified water would be

Q.22

The useful work done during the reaction

$\mathrm{Ag_{(s)} + \frac{1}{2}Cl_{2(\ g)} \rightarrow AgCl_{(s)}}$

would be _____ (Report your answer as closest integer in kJ)

(Given $\mathrm{E^0_{Cl_2/Cl^-}} = 1.36\ \mathrm{V}, \mathrm{E^0_{AgCl/Ag/Cl^-}} = 0.220\ \mathrm{V}, \mathrm{P_{Cl_2}} = 1\ \mathrm{atm}$, and $\mathrm{T} = 298\ \mathrm{K}$)

Q.23

A $100\ \mathrm{watt}, 110\ \mathrm{volt}$ lamp is connected in series with an electrolytic cell containing cadmium sulphate solution. What mass of cadmium will be deposited by the current flowing for $10\ \mathrm{hours}$? (Given, Atomic mass of $\mathrm{Cd} = 112.4$)(round off to the nearest integer)

Q.24

The conductivity of the saturated solution of some bivalent salt XY is $3.06 \times 10^{-6}\ \mathrm{ohm^{-1}\ cm^{-1}}$ and its

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equivalent conductivity is $1.53\ \text{ohm}^{-1}\ \text{cm}^2\text{equivalent}^{-1}$. The value of K_{sp} of XY is $p \times 10^{-q}$, then the value of $p + q$ is?

Q.25

If the Pb^{2+} concentration is maintained at $1.0\ M$ then the $\left[Cu^{2+}\right]$ when the cell potential drops to zero is 1×10^{-x}, the value of x is?
$E^{o}_{cell} = 0.473\ V, Pb\left|Pb^{2+}(1.0M)\right|\left|Cu^{2+}\left(1.0 \times 10^{-4}M\right)\right|Cu(s)$

Q.26

The current in a given wire is $1.8\ A$. The number of coulombs that flow in 1.36 minutes will be

Q.27

Resistance of a conductivity cell filled with a solution of an electrolyte of concentration $0.1\ M$ is $100\ \Omega$. The conductivity of this solution is $1.29\ S\ m^{-1}$. Resistance of the same cell when filled with $0.2\ M$ of the same solution is $520\ \Omega$. The molar conductivity of $0.02\ M$ solution of the electrolyte will be $p \times 10^{-5}\ S\ m^2\ mol^{-1}$, then the value of p is

Q.28

An ion M^{n+} (molar mass =30) discharged at cathode according to the following reaction
$M^{n+} + C_2O_4^{2-} \rightarrow CO_2 + M$
$10\ ml$ of $0.1\ M$ solution of M is converted into M^{n+} by $20\ ml$ of $0.03\ M$ solution of $KMnO_4$ in acidic medium. The weight of 'M' can be produced when $3\ ampere$ current is passed through the solution for $9.65\ hours$ is (round off to the nearest integer)

Q.29

$0.1\ M$ aqueous solution of a dibasic acid is diluted such that its equivalent conductance is five times. Hence, final concentration is k then $100 \times k$ is

Q.30

During discharging of the lead-storage acid battery the following reaction takes place :
$Pb(s) + PbO_2(s) + 2H_2SO_4 \rightarrow 2PbSO_4(s) + 2H_2O$
If $2.5\ amp$ of current is drawn for $965\ minutes$, H_2SO_4 consumed is k, then the value of $2k$ is

Answers & Solutions

Q.1 Answer:

$19.3 \times 10^4\ s$

Solution:

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$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

For 0.01 **mole** H_2 0.02 **mole of electrons are consumed**

charge required $= 0.02 \times 96500\ C = i \times t$

Time required $= \frac{0.02 \times 96500}{10 \times 10^{-3}} = 19.3 \times 10^4\ s$

Q.2 Answer:

$2.05 \times 10^{-5} M$

Solution:

Molar conductivity at infinite dilution of KCl **is**

$\Lambda_m^\infty (KCl) = \lambda_m^\infty (K^+) + \lambda_m^\infty (Cl^-)$

$= (73.52 + 76.34) \Omega^{-1}\ cm^2\ mol^{-1}$

$= 149.86\ \Omega^- cm^2\ mol^{-1}$

Conductivity of $0.01\ M\ KCl$ **is**

$K_1 = \Lambda_m^\infty c = \left(149.86\ \Omega^{-1}\ cm^2\ mol^{-1}\right)$

$\left(0.01 \times 10^{-3}\ mol\ cm^{-3}\right) = 1.4986 \times 10^{-3} \Omega^{-1}\ cm^{-1}$

Cell constant is

$K = k_1 R = \left(1.4986 \times 10^{-3} \Omega^{-1}\ cm^{-1}\right)(525\ \Omega) = 0.7868\ cm^{-1}$

Conductivity of $0.1\ M\ NH_4OH$ **is**

$\Lambda_c\ (NH_4OH) = \frac{k_2}{c} = \frac{3.876 \times 10^{-4} \Omega^{-1}\ cm^{-1}}{0.1 \times 10^{-3}\ mol\ cm^{-3}}$

Molar conductivity of $0.1\ M\ NH_4OH$ **is**

$\Lambda_c\ (NH_4OH) = \frac{k_2}{c} = \frac{3.876 \times 10^{-4} \Omega^{-1}\ cm^{-1}}{0.1 \times 10^{-3}\ mol\ cm^{-3}}$

$= 3.876\ \Omega^{-1}\ cm^2\ mol^{-1}$

Molar conductivity of NH_4OH **at infinite dilution is**

$\Lambda_m^\infty\ (NH_4OH) = \lambda_m^\infty\ (NH_4^+) + \lambda_m^\infty\ (OH^-)$

$= (73.4 + 197.6) \Omega^{-1}\ cm^2\ mol^{-1}$

$= 271.0\ \Omega^{-1}\ cm^2\ mol^{-1}$

Degree of dissociation of NH_4OH **is**

$\alpha = \frac{\Lambda_c}{\Lambda_m^\infty} = \frac{3.876 \Omega^{-1}\ cm^2\ mol^{-1}}{271.0 \Omega^{-1}\ cm^2\ mol^{-1}} = 0.0143$

Equilibrium constant of NH_4OH **is**

$K = \frac{[NH_4^+][OH^-]}{[NH_4OH]} = \frac{(c\alpha)(c\alpha)}{c(1-\alpha)} \simeq c\alpha^2$

$= (0.1M)(0.0143)^2 = 2.05 \times 10^{-5}\ M$

Q.3 Answer:

$0.52\ V$

Solution:

Reactions

$Cu^{2+} + 2e^- \rightarrow Cu \qquad \Delta G^0 = -nFE^0$

$Cu^+ \rightarrow Cu^{2+} + e^-$

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$\Delta G_{I}^{0} = -2\ F \times 0.34 = 0.68\ F$

$\Delta G_{II}^{0} = F \times 0.16 = 0.16\ F$

$\Delta G_{I}^{0} = -0.68\ F$

$\Delta G_{II}^{0} = 0.16\ F$

Adding, we get

$Cu^{+} + e^{-} \rightarrow Cu \quad \Delta G_{III}^{0} = \Delta G_{I}^{0} + \Delta G_{II}^{0}$

$= 0.52\ F$

$= -FE^{0}$

$\therefore E^{\circ} = 0.52\ V$

Q.4 Answer:

12.13

Solution:

After electrolysis, aqueous $NaCl$ is converted into aqueous $NaOH$.

The quantity of electricity passed $= \frac{0.250 \times 35 \times 60}{96500}\ F = 5.44 \times 10^{-3}\ F$

The number of equivalents of OH^{-} ion formed $= 5.44 \times 10^{-3}$

$\therefore$ **Morality of** $NaOH =$

$\frac{5.44 \times 10^{-3}}{0.4\ L} = 1.36 \times 10^{-2}$

$\therefore pOH = -\log\left(1.36 \times 10^{-2}\right) = 1.87$

$\therefore pH = 12.13$

Q.5 Answer:

$6.43 \times 10^{5} C$

Solution:

$ClO_{4}^{-} + 2H^{+} + 2e^{-} \rightarrow ClO_{3}^{-} + H_{2}O$

Number of equivalents of

$NaClO_{4} = \frac{245}{61.25} = 4 = 4\ F$

No. of Faradays $= \frac{4 \times 100}{60} = 6.67\ F$

$= 6.43 \times 10^{5} C$

Q.6 Answer:

MnO_{4}^{-} can be used in aqueous HCl

Solution:

The reaction between MnO_{4}^{-} and HCl may be represented as follows:

$2MnO_{4}^{-}(aq) + 16H^{+} + 10Cl^{-} \rightarrow$

$2Mn^{2+}(aq) + 8H_{2}O(l) + 5Cl_{2}(\ g)$

Thus, on the basis of this reaction following electrochemical cell will be represented

$Pt, Cl(g)(1atm)\left|Cl^{-}(aq)\right|\left|MnO_{4}^{-}(g)\right|\ Mn^{2+}(aq)$

Hence, $E_{cell}^{0} = E_{cathode}^{0} - E_{anode}^{0}$

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From given data $E^0_{\text{cell}} = 1.51 - 1.40 = 0.11\ \text{V}$

E° cell is positive, hence ΔG° is negative. Thus, above cell reaction is feasible but MnO_4^- ion can oxidise, Fe^{2+} to Fe^{3+} and Cl^- to Cl_2 in aqueous medium also. Therefore, for quantitative estimation of aqueous $\text{Fe}\left(\text{NO}_3\right)_2$ it is not a suitable reagent.

Q.7 Answer:

$\wedge^\circ_{\text{NaCl}}$

Solution:

$$\wedge^{\text{o}}_{\text{HOAc}} = \wedge^{\text{o}}_{\text{H+}} + \bigwedge^{\text{o}}_{\text{AcO}-}$$
$$= \left(\wedge^{\text{o}}_{\text{H}^+} + \wedge^{\text{o}}_{\text{Cl}^-}\right) + \left(\wedge^{\text{o}}_{\text{AcO}-} + \bigwedge_{\text{Na}^+}\right)\left(-\wedge^{\text{o}}_{\text{Cl}-} - \wedge^{\text{oA}}_{\text{Na+}}\right)$$
$$= \left(\wedge^{\text{o}}_{\text{HCl}}\right) + \left(\wedge^{\text{o}}_{\text{NaOAc}}\right) - \left(\wedge^{\text{o}}_{\text{NaCl}}\right)$$

Q.8 Answer:

HgCl_2

Solution:

HgCl_2 has corrosive action. It is highly poisonous. It sublimes on heating. It is, therefore, known as corrosive sublimate.

Q.9 Answer:

1

Solution:

1) $2\text{Fe}^{+3} + \text{Fe} \rightarrow 2\text{Fe}^{++} + \text{Fe}^{++}$

$\text{E}^0_{\text{reaction}} = \text{E}^0_{\text{O.A}} - \text{E}^0_{\text{R.A}}$

$= 0.77 - (-0.44)$

$= +1.21\ \text{V}$

E^0 is positive so ΔG is negative

$\therefore$ Spontaneous reaction

2) $\text{Hg}_2^{++} \rightarrow \text{Hg}^{++} + \text{Hg}$

$\text{E}^0_{\text{redox Reaction}} = \text{E}^0_{\text{O.A}} - \text{E}^0_{\text{R.A}}$

$= 0.85 - 0.92$

$= -0.07$

E^0 is negative so ΔG is positive

$\therefore$ nonspontaneous reaction

3) $3\text{AgCl} + \text{NO} + 2\text{H}_2\text{O} \rightarrow 3\text{Ag} + 3\text{Cl}^- + \text{NO}_3^- + 4\text{H}^+$

$\text{E}^0_{\text{redox Reaction}} = \text{E}^0_{\text{O.A}} - \text{E}^0_{\text{R.A}} = 0.22 - 0.96 = -074.\text{V}$

E^0 is negative so ΔG is positive

$\therefore$ non-spontaneous reaction

Q.10 Answer:

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Mg

Solution:

The reduction potential of Mg is less than that of water $\left(E^0 = -0.83\ V\right)$. Hence their ions in the aqueous solution cannot be reduced instead water will be reduced

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

Q.11 Answer:

$2x$

Solution:

$2H_2O + 2e^- \rightarrow H_2 + 2OH^- \ldots\ldots I$

$2H_2O \rightarrow O_2 + 4H^+ + 4e^- \ldots\ldots II$

From I : 1 mol of $H_2 = 22.4\ L H_2$ is obtained by 2 units

x L of $H_2 = \frac{2x}{22.4}$ units costing Rs. x

Thus, 1 unit costs = Rs. 11.2

from II : 22.4 L O_2 at NTP costs = 4 units

$= 4 \times 11.2 Rs$

x L costs = $2x$

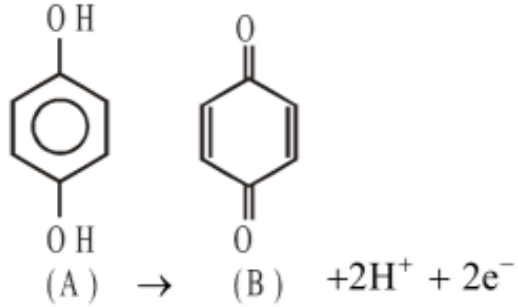
Q.12 Answer:

$1.42\ V$

Solution:

In this electrode

$[A] = [B]$ **in quinhydrone electrode**



Hence,

$Q = [H^+]^2$

$E = E^\circ - \frac{0.0591}{2} \log [H^+]^2 = E^0 - 0.0591 \log [H^+] = E^\circ + 0.0591 pH = 1.30 + 0.0591 \times 2 \approx 1.42\ V$

Q.13 Answer:

$\frac{[Zn^{2+}]}{[Cu^{2+}]} = 0.01$

Solution:

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$E_{cel} = E^{o}_{cell} - \frac{0.0591}{2} \log \left[\frac{Zn^{2+}}{Cu^{2+}}\right]$

from graph $OA = E^{o}_{cell} = 1.01\ V$

$E_{cell} = 1.159\ V$

Substituting all values

$\frac{[Zn^{2+}]}{[Cu^{2+}]} = 10^{-2}\ M$

$= 0.01$

Q.14 Answer:

Potentiometer

Solution:

The standard electrode potential is measured by the potentiometer. It is the potential difference when no current is flowing in the circuit.

Q.15 Answer:

Electrodes made of calcium-containing lead alloy

Solution:

Calcium-tin-lead alloys are used for making grids for maintenance-free batteries. The calcium content in such alloys for positive grids has varied generally from about 0.06 to about 0.1% by weight of the alloy while the tin has generally ranged from about 0.1 up to 0.8% and even more. More typically, the calcium content in such alloys when used for making maintenance-free battery grids has been at least about 0.08% by weight or more.

Q.16 Answer:

velocity of both K^{+} and NO_3^{-} are nearly the same

Solution:

For a good salt bridge, the velocities of ions should be same. If the velocities are not same, then the ions would not be able to neutralize the charge equally in both half cells. Consequently, an opposite force would be developed that restricts the flow of current. For KNO_3, the velocities of K^{+} and NO_3 - are nearly same. Therefore, it is used to make salt bridges.

Q.17 Answer:

$2Cr^{3+} + 7H_2O \rightarrow Cr_2O_7^{2-} + 14H^{+}$

Solution:

Oxidation always occurs at anode.

Q.18 Answer:

$Pt/H_2(\ g)HCl(soln)||AgCl(s)/Ag$

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Solution:

H_2 **undergoes oxidation and** $AgCl\ (Ag^+)$ **undergoes reduction.**

Q.19 Answer:

Y will oxidize X and not Z

Solution:

The tendency to gain electron is in the order $Z > Y > X$

Thus $Y + e^- \rightarrow Y^-; \quad X \rightarrow X^+ + e^-.$

Q.20 Answer:

0.59 V

Solution:

$E_{OP} = E^\circ_{OP} - \frac{0.059}{1} \log \frac{[H^+]}{P_{H_2}}$

$\therefore [H^+] = 10^{-10}; P_{H_2} = 1$ **atm** $; E_{OP} = 0.59V.$

Q.21 Answer:

—

Solution:

$2H_2O \rightarrow 2H_{2(g)} + O_{2(g)}$

$2x \qquad x$

$\therefore 3x = 0.168$

$\therefore x = 0.056$ L

$V_{H_2} = 2x = 0.112$ L, $V_{O_2} = x = 0.056$ L

11.2 L **of** H_2 **at** STP $\equiv 1$ F

0.112 L **of** H_2 **at** STP $\equiv 0.01$ F

0.056 L **of** O_2 **at** STP $= 0.01$ F

$\therefore$ **The amount of electricity passed**

$= 0.01$ F $= 965$C

Q.22 Answer:

—

Solution:

For the cell reaction

$Ag_{(s)} + \frac{1}{2}Cl_{2(g)} \rightarrow AgCl_{(s)}$

$E^0 = -1.14$ V

$\therefore$ **Useful work** $= -W_{max} = -nFE$

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$= (-1) \times (-1.14) \times 96500 \times 10^{-3}$ KJ
$= 110$ KJ mol^{-1}

Q.23 Answer:

—

Solution:

$P = 100$ **watt**
$V = 110$ **volt**
Time $= 10$ hrs $= 10 \times 60 \times 60$
$= 3.6 \times 10^4$sec
$P = V \times i$
$i = \frac{P}{V} = \frac{100}{110}$
$W = Zit = \frac{Eit}{96500}$
$= \frac{112.4}{2} \times \frac{100}{110} \times \frac{3.6 \times 10^4}{96500} = 19.05$ gm
Approx $= 19$ gm

Q.24 Answer:

—

Solution:

$\wedge_{eq} = \frac{K \times 1000}{N}$
or,
$N = \frac{K \times 1000}{\wedge_{eq}} = \frac{3.06 \times 10^{-6} \times 10^3}{1.53} = 2 \times 10^{-3}$ N
Molarity of X $Y = 2 \times 10^{-3} \times \frac{1}{2} = 1 \times 10^{-3}$
$K_{sp} = \left[X^{2+}\right]\left[Y^{-2}\right] = \left(1 \times 10^{-3}\right)\left(1 \times 10^{-3}\right) = 1 \times 10^{-6}$

Q.25 Answer:

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Solution:

$$Pb \rightarrow Pb^{2+} + 2e^-$$
$$Cu^{2+} + 2e^- \rightarrow Cu$$
$$\overline{Pb + Cu^{2+} \rightarrow Pb^{2+} + Cu}$$

$K = \frac{\left[Pb^{2+}\right]}{\left[Cu^{2+}\right]} = \frac{1}{\left[Cu^{2+}\right]}$
$\therefore \quad E = E^\circ - \frac{0.0591}{2} \log K$
$= E^\circ + \frac{0.0591}{2} \log \left[Cu^{2+}\right] 0 = 0.473 + \frac{0.0591}{2} \log \left[Cu^{2+}\right]$

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$\log\left[Cu^{2+}\right] = -16$
$\therefore \left[Cu^{2+}\right] = 1 \times 10^{-16}\ M$

Q.26 Answer:

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Solution:

$Q = I \times t; 1.8 \times 1.36 \times 60 = 147\ C.$

Q.27 Answer:

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Solution:

Cell constant =
$\frac{\text{conductivity(I)}}{\text{conductance(I)}} = \frac{\text{conductivity(II)}}{\text{conductance(II)}}$
Conductivity $(II) = \frac{129}{520} sm^{-1}$
$0.2M = 200\ mol\ m^{-3}$
Molar conductivity of $0.2M$ **solution**
$= \frac{129/520}{200}$
$= 12.4 \times 10^{-4}\ sm^2\ mol^{-1}$
From the given data it is not possible to calculate molar conductivity of $0.02\ M$ **solution exactly - However, it is sure that its molar conductivity will be slightly greater than** 12.4×10^{-4} $sm^2\ mol^{-1}$**. Assuming linear inverse variation of molar conductivity with concentration we can say that**
$\therefore$ **Molar conductivity of** $0.02\ M$ **solution**
$= 1.24 \times 10^{-3} \times 10$
$= 124 \times 10^{-4}\ s\ m^2\ mol^{-1}$

Q.28 Answer:

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Solution:

On the basis of titration gm **eq of**
$M^{2+} =$ gm **meq of** $KMnO_4$
$'n' \times 0.1 \times 10 \times 10^{-3} = 20 \times 10^{-3} \times 0.03 \times 5$
$\therefore \quad n'_{factor} = 3$ $\therefore$ **eq. mass of** $M^{2+} = \frac{\text{M.mass}}{3}$
$= \frac{30}{3} = 10\ g$
$\therefore \quad \frac{W}{10} = \frac{3 \times 9.65 \times 60 \times 60}{96500}$
$W = 10.8gm$

Q.29 Answer:

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Solution:

$0.1\ \mathrm{M}$ **dibasic acid** $= 0.2\ \mathrm{N}$

$\Lambda_{\mathrm{C}} = \frac{1000 \times \text{sp. conductance}}{\text{conc.(Normality)}} = \frac{1000\ \mathrm{S}}{0.2}$

$\Lambda'_{\mathrm{C}} = \frac{1000 \times \mathrm{S}}{\mathrm{N}'}$

$\Lambda'_{\mathrm{C}} = 5\Lambda_{\mathrm{C}}$

$\frac{1000\ \mathrm{S}}{\mathrm{N}'} = 5 \times \frac{1000\ \mathrm{S}}{0.2}$

$\frac{1}{\mathrm{N}'} = \frac{5}{0.2}$

$\mathrm{N}' = \frac{0.2}{5} = 0.04\ \mathrm{N}$

$\mathrm{k} = 0.04$

$100 \times 0.04 = 4$

Q.30 Answer:

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Solution:

Oxidation : $\mathrm{Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(\ s) + 2e^-}$

Reduction :

$\mathrm{PbO_2(\ s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \rightarrow PbSO_4(\ s) + 2H_2O(}l\mathrm{)}$

During discharging $\mathrm{H_2SO_4}$ **consumed**

$\mathrm{W = zit}$

Mass of $\mathrm{H_2SO_4} = 2\ \mathrm{mol} = 2 \times 98\ \mathrm{g}$

$\therefore \mathrm{Z} = \frac{\text{mass of } \mathrm{H_2SO_4}}{\text{electrons} \times 96500}$

$\therefore \mathrm{w} = \frac{2 \times 98 \times 2.5 \times 965 \times 60}{2 \times 96500}$

$= 147\ \mathrm{g} = 1.5\ \mathrm{mol} = \mathrm{k}$

$2\mathrm{k} = 3$

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